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AD3811 – PROJECT WORK
ZEROTH REVIEW

A Multi-Agent Reinforcement Learning Architecture with Independent Sentiment-Anomaly Veto for Pump-and-Dump-Resistant Trading

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Abstract

Automated trading systems built on price and volume signals cannot distinguish a genuine, fundamentals-driven price spike from a coordinated pump-and-dump, since both surface identically as "price moving up fast." Recent multi-agent reinforcement learning (MARL) architectures for trading and portfolio management — including Agentic AI ensembles that combine LSTM, Transformer, and XGBoost prediction with a FinBERT-based Sentiment Agent resolved by majority vote, and modular systems such as MSPM, which route FinBERT sentiment into asset-specific Deep Q-Network agents whose outputs feed a PPO-based portfolio allocator — already incorporate sentiment analysis. In every such system, however, the sentiment signal is treated as one more input feature blended into the decision with no explicit way to distinguish routine sentiment from statistically abnormal hype, and neither system is evaluated against labeled manipulation events.

This project proposes a Multi-Agent Reinforcement Learning Architecture with an Independent Sentiment-Anomaly Veto for pump-and-dump-resistant crypto trading. The system comprises an Analysis Agent that proposes buy, sell, or hold actions from price-action and technical indicators via a reinforcement-learning policy; a Risk Agent that filters proposals against position-sizing and volatility limits; and a Sentiment/Hype Deflator Agent that scores live news and social media with FinBERT and flags statistically abnormal emotional intensity relative to an asset's historical baseline. Unlike prior work, the Deflator Agent sits outside the voting loop with explicit override authority to force a hedge or reversal whenever hype is anomalous, regardless of what the other agents vote.

The system will be backtested on historical crypto price, news, and social data spanning documented pump-and-dump events, comparing a price-only baseline, a sentiment-as-feature configuration replicating existing literature, and the proposed sentiment-as-veto configuration, evaluated on Sharpe ratio, maximum drawdown, and manipulation-detection precision and recall.

Keywords: *Multi-Agent Reinforcement Learning, FinBERT, Sentiment Anomaly Detection, Pump-and-Dump Detection, Algorithmic Trading, Override/Veto Mechanism, Crypto Market Manipulation*

References

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